

Elisa Tianqi Duan

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EDUCATION

Northwestern University

Bachelor of Arts in Data Science

Bachelor of Science in Industrial Engineering & Management Sciences (IEMS)

GPA: 3.86/4.00

Relevant Coursework: Data Structures and Algorithms, Data Management and Information Processing, Data Science with Python, Statistical Theory and Methods, Probability, Linear Algebra, Optimization, Stochastic Modeling

Evanston, IL

Expected June 2027

SKILLS

- **Programming Languages:** Python, SQL, R, MATLAB, Racket, AMPL, HTML
- **Libraries:** pandas, NumPy, scikit-learn, matplotlib, seaborn, statsmodels
- **Tools & Technologies:** PostgreSQL, Snowflake, MongoDB, Git, GitHub, Jupyter Notebook, VS Code, RStudio, Looker Studio
- **Languages:** English (Fluent), Chinese (Fluent)

PROFESSIONAL EXPERIENCE

Strategy & Operations Intern, Enova International | Chicago, Illinois

June 2026

- Developing complex Snowflake and PostgreSQL queries to integrate loan, customer, and bank account history data into an event-level dataset **supporting analysis of customer post-funding bank account update behavior**
- Analyzing fraud, payment return, and repayment outcomes across products and lenders **to inform risk-based verification policies that balance fraud prevention with customer experience**

Data Science Intern, IDX Exchange | Remote

September 2025 – December 2025

- Collaborated on exploratory data analysis (EDA), feature engineering, and preprocessing on **178K+ real estate records**, reducing 80 candidate variables to 23 high-value predictors across seven predictive housing price models
- Developed and evaluated a LightGBM regression model using California Regional Multiple Listing Service (CRMLS) data **to improve residential property price forecasting**
- Benchmarked multiple machine learning models using R^2 , MAPE, and MdAPE, identifying CatBoost as the **top-performing model** with testing R^2 of 0.889 and MdAPE of 8.03%

Data Science Intern, Metropolitan Chicago Data-science Corps | Chicago, Illinois

June 2025 – August 2025

- Designed an 8-table PostgreSQL relational database using Supabase and a responsive HTML/CSS/JavaScript web interface via GitHub Pages **to streamline the archival, search, and retrieval of historical seismic records**
- Developed searchable data-entry and visualization interfaces, integrating frontend forms with backend **APIs to support structured archival and retrieval workflows for historical seismogram records**
- Partnered with Born Physical and Studied Digitally Consortium to digitize and annotate historic seismic microfilm archives using PowerScan200 and Label Studio, extracting metadata to support long-term seismic research accessibility

Teaching Assistant, Northwestern University Computer Science Department | Evanston, Illinois

March 2025 – Present

- Mentored 100+ students in CS 217: Data Management and Information Processing, teaching SQL and relational database management using SQLiteStudio through database creation, data importing, and query development using real-world datasets
- Delivered 1:1 code walkthroughs and debugging support to strengthen student proficiency in SQL and database design

Operations & Finance Intern, MATTER Health | Chicago, Illinois

June 2025 – August 2025

- Built automated ETL-style workflows using Google Apps Script to centralize rolling 12 months of data for room bookings, employee attendance, and visitor activity while developing 3 real-time Looker Studio dashboards **to improve operational visibility and space utilization tracking**
- Consolidated and analyzed fiscal-year QuickBooks P&L data in Excel for audit compliance reporting and streamlined a Google Sheets commission-tracking system to improve compensation accuracy and financial reporting visibility

PROJECTS

Adidas Sales Analysis | Python

September 2025 – June 2026

- Analyzed 9,500+ Adidas sales records **to identify key drivers of profitability across retailers, products, and sales channels**; visualized data using pandas, NumPy, matplotlib, and seaborn to create 8 business-focused visualizations
- Built 2 predictive models in scikit-learn and statsmodels using 10+ predictors to forecast operating profit; conducted regression analysis and statistical inference using R^2 , RMSE, p-values, confidence intervals, and joint F-tests, achieving an R^2 of 0.914 and identifying statistically significant profitability drivers

Chicago Crime Data Analysis | Python

January 2025 – March 2025

- Explored 3 Chicago crime datasets spanning 1991–Present using pandas, matplotlib, and seaborn **to examine demographic disparities in violent crime victimization across race, age, and sex**
- Cleaned, merged, and visualized large-scale crime data to identify victimization trends across demographic groups